

Literature Status: The Shift-Invariant Predual Question for $\ell^1(\mathbb{Z})$

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Status: literature already answered. The concrete unique-predual question in arXiv:math/0604372 is answered negatively by the separate paper arXiv:1101.5696. There are continuum many distinct shift-invariant preduals of $\ell^1(\mathbb{Z})$, hence continuum many compatible weak-star topologies, all different from the canonical one.

1 Original question

Daws asks in Section 4, PDF page 13, of *Dual Banach algebras: representations and injectivity* [1] whether $\ell^1(\mathbb{Z})$ has a unique predual compatible with convolution. His equivalent concrete formulation is:

Let $X \subseteq \ell^\infty(\mathbb{Z})$ be a shift-invariant subspace such that X' is naturally identified with $\ell^1(\mathbb{Z})$. Is $X = c_0(\mathbb{Z})$?

The compatibility condition means that convolution on $\ell^1(\mathbb{Z})$ is separately weak-star continuous for the duality induced by X .

2 Separate later answer

Daws, Haydon, Schlumprecht, and White explicitly cite the source paper and state on PDF page 2 of *Shift invariant preduals of $\ell_1(\mathbb{Z})$* that their results answer the question negatively [2]. Proposition 2.3 identifies compatible preduals with concrete shift-invariant subspaces of $\ell^\infty(\mathbb{Z})$. Theorem 3.4 and Corollary 3.5, PDF pages 9–10, construct preduals $F^{(\lambda)}$ for every $|\lambda| > 1$ and prove that they induce continuum many distinct weak-star topologies on $\ell^1(\mathbb{Z})$, each different from the topology induced by $c_0(\mathbb{Z})$.

Therefore $X = c_0(\mathbb{Z})$ is false in general, and the compatible predual is not unique. The same paper later gives examples not even isomorphic to c_0 as Banach spaces.

3 Scope and status audit

This completely answers the source's exact convolution-compatible unique predual question. It does not classify all shift-invariant preduals. Nor does it refute the stronger result, discussed in the answer paper, that the canonical predual is unique if both convolution and the natural coproduct are required to be weak-star continuous.

The supporting authors explicitly knew they were answering Daws’s question: their introduction names the first author’s question, cites the source as reference [12], and says that the results answer it negatively. A bounded search through 11 August 2026 checked the run indexes, both arXiv ids, the exact question wording, title/author searches, the source citation in the answer paper, and the journal metadata. This is an exact literature answer, not an original mathematical result.

References

- [1] M. Daws, *Dual Banach algebras: representations and injectivity*, Studia Math. **178** (2007), 231–275; arXiv:math/0604372; DOI 10.4064/sm178-3-3.
- [2] M. Daws, R. Haydon, Th. Schlumprecht, and S. White, *Shift invariant preduals of $\ell_1(\mathbb{Z})$* , Israel J. Math. **192** (2012), 541–585; arXiv:1101.5696; DOI 10.1007/s11856-012-0040-1.